Chapter 5: Ethernet



Introduction to Networks

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(Logical Link Control) LLC and (Medium Access Control) MAC Sublayers

Ethernet

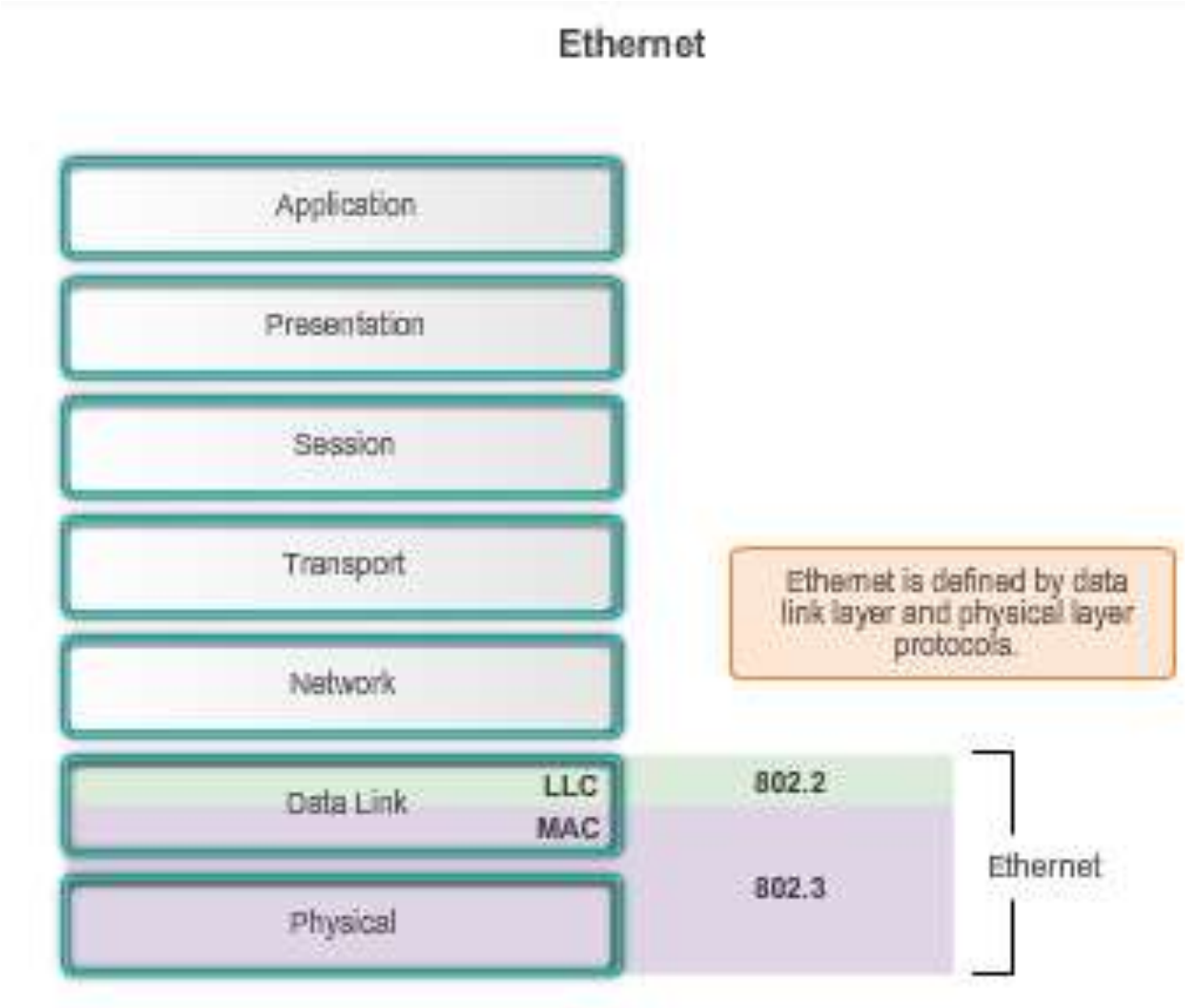
- One of the most widely used LAN technologies
- Operates in the data link layer and the physical layer
- Family of networking technologies that are defined in the IEEE 802.2 and 802.3 standards
- Supports data bandwidths of 10, 100, 1000, 10,000, 40,000, and 100,000 Mbps (100 Gbps)

Ethernet Standards

- Define Layer 2 protocols and Layer 1 technologies
- Two separate sub layers of the data link layer to operate – Logical link control (LLC) and the MAC sublayers

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LLC and MAC Sublayers (cont.)



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LLC and MAC Sublayers (cont.)

LLC

- Handles communication between upper and lower layers.
- Takes the network protocol data and adds control information to help deliver the packet to the destination.

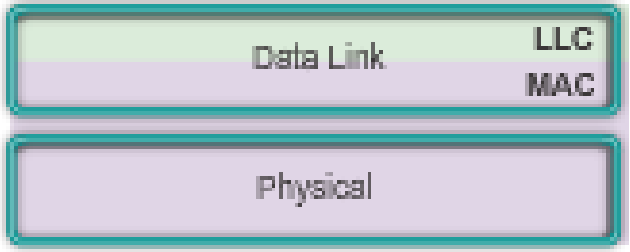
MAC

- Constitutes the lower sublayer of the data link layer.
- Implemented by hardware, typically in the computer NIC.
- Two primary responsibilities:
  - Data encapsulation
  - Media access control



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MAC Sublayer



Data encapsulation

- Frame assembly before transmission and frame disassembly upon reception of a frame.
- MAC layer adds a header and trailer to the network layer (Protocol Data Unit) PDU.

Provides three primary functions:

- **Frame delimiting** – Identifies a group of bits that make up a frame, synchronization between the transmitting and receiving nodes.
- **Addressing** – Each Ethernet header added in the frame contains the physical address (MAC address) that enables a frame to be delivered to a destination node.
- **Error detection** – Each Ethernet frame contains a trailer with a cyclic redundancy check (CRC) of the frame contents.

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MAC Sublayer (cont.)

MAC

- Responsible for the placement of frames on the media and the removal of frames from the media
- Communicates directly with the physical layer
- If multiple devices on a single medium attempt to forward data simultaneously, the data will collide resulting in corrupted, unusable data
- Ethernet provides a method for controlling how the nodes share access through the use of Carrier Sense Multiple Access (CSMA) technology



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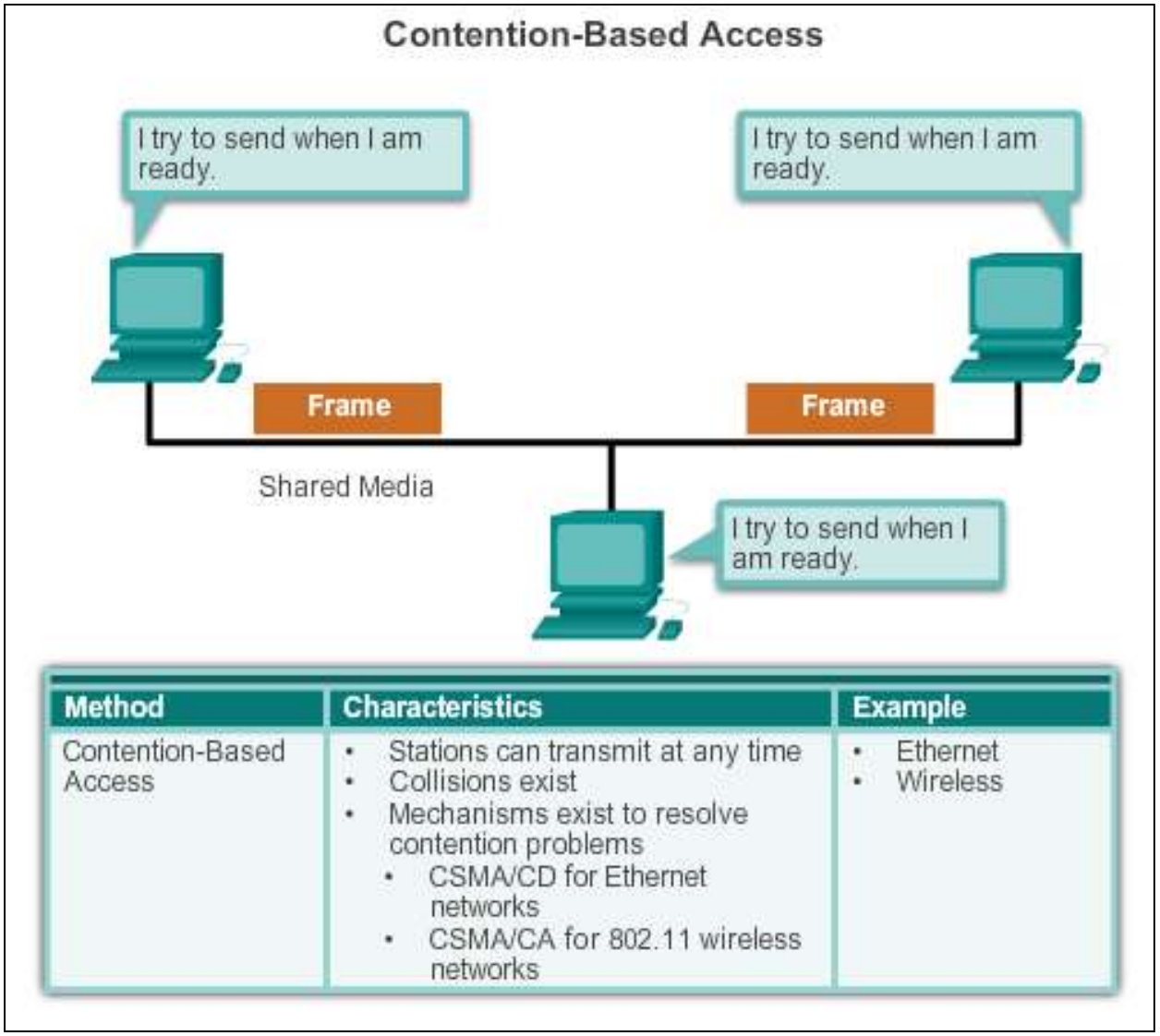
Media Access Control

Carrier Sense Multiple Access (CSMA) process

- Used to first detect if the media is carrying a signal
- If no carrier signal is detected, the device transmits its data
- If two devices transmit at the same time - data collision

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Media Access Control (cont.)

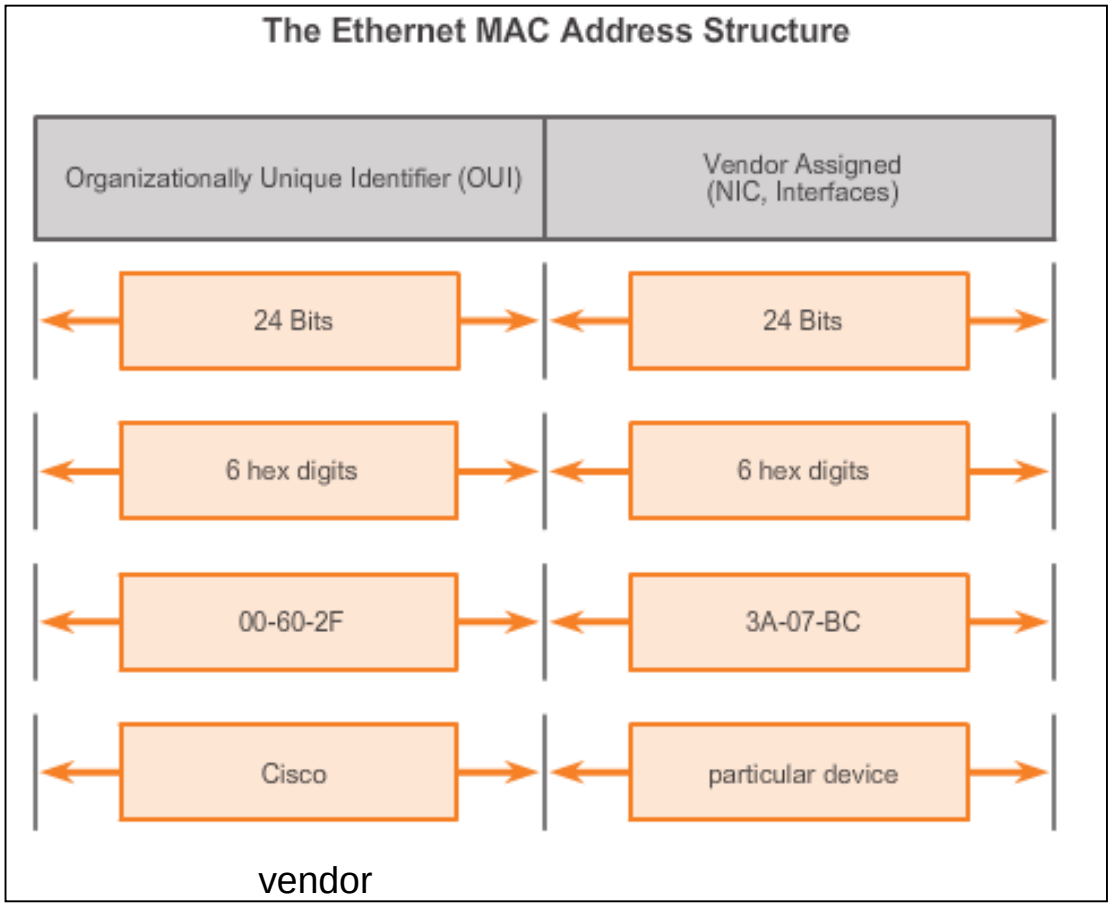


Method	Characteristics	Example
Contention-Based Access	• Stations can transmit at any time • Collisions exist • Mechanisms exist to resolve contention problems • CSMA/CD for Ethernet networks • CSMA/CA for 802.11 wireless networks	• Ethernet • Wireless

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MAC Address: Ethernet Identity

- Layer 2 Ethernet MAC address is a 48-bit binary value expressed as 12 hexadecimal digits.
- IEEE requires a vendor to follow these rules:
  - Must use that vendor's assigned OUI as the first 3 bytes.
  - All MAC addresses with the same OUI must be assigned a unique value in the last 3 bytes.



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Frame Processing

- MAC addresses assigned to workstations, servers, printers, switches, and routers.
- Example MACs:
  - 00-05-9A-3C-78-00
  - 00:05:9A:3C:78:00
  - 0005.9A3C.7800.
- When a device is forwarding a message to an Ethernet network, attaches header information to the packet, contains the source and destination MAC address.
- Each NIC views information to see if the destination MAC address in the frame matches the device's physical MAC address stored in RAM.
- No match, the device discards the frame.
- Matches the destination MAC of the frame, the NIC passes the frame up the OSI layers, where the de-encapsulation process takes place.

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Ethernet Frame Size

- Ethernet II and IEEE 802.3 standards define the minimum frame size as 64 bytes and the maximum as 1518 bytes
- Less than 64 bytes in length is considered a "collision fragment" or "runt frame"
- If size of a transmitted frame is less than the minimum or greater than the maximum, the receiving device drops the frame
- At the physical layer, different versions of Ethernet vary in their method for detecting and placing data on the media

IEEE 802.3

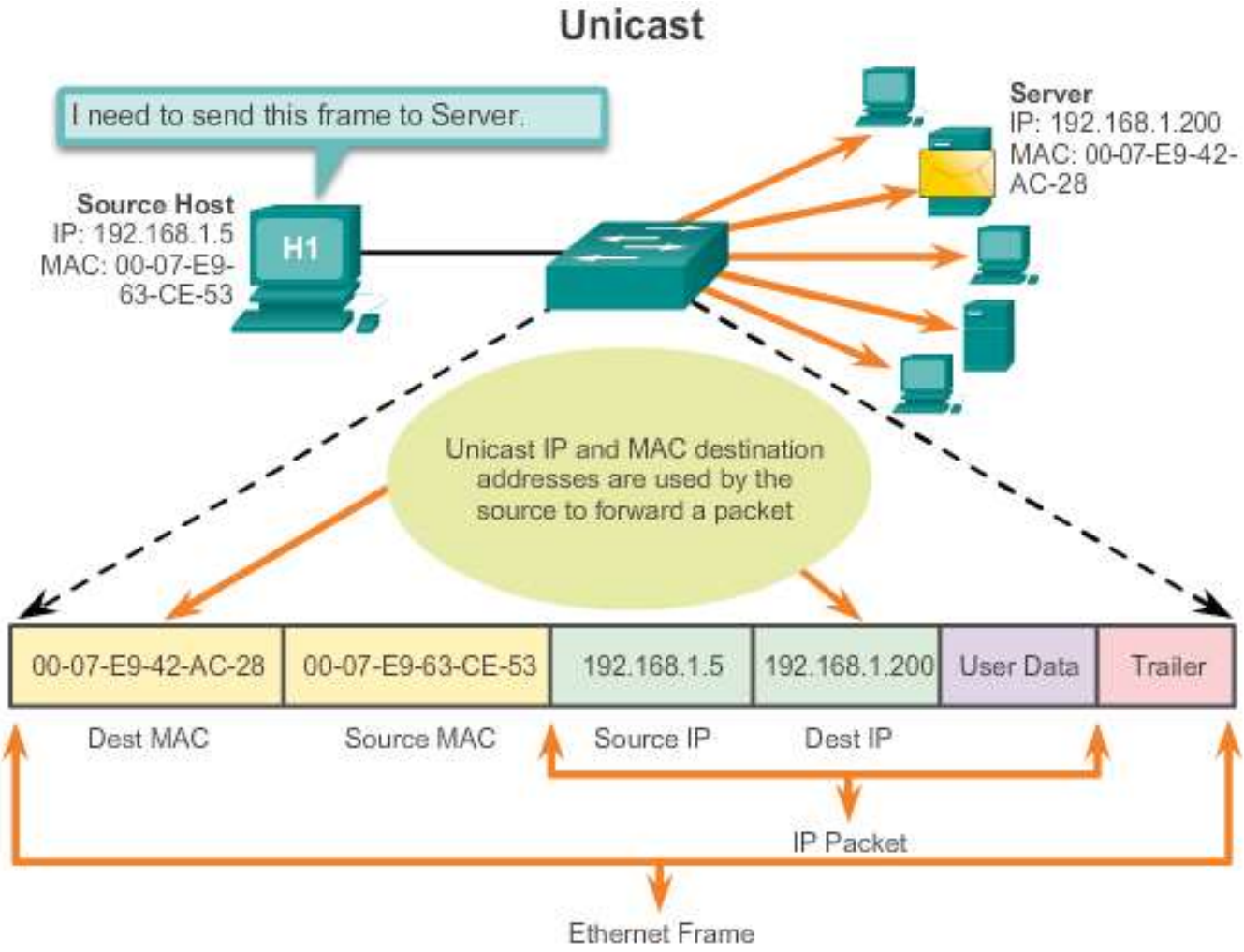
7	1	6	6	2	46 to 1500	4
Preamble	Start of Frame Delimiter	Destination Address	Source Address	Length	802.2 Header and Data	Frame Check Sequence

Ethernet II

8	6	6	2	46 to 1500	4
Preamble	Destination Address	Source Address	Type	Data	Frame Check Sequence

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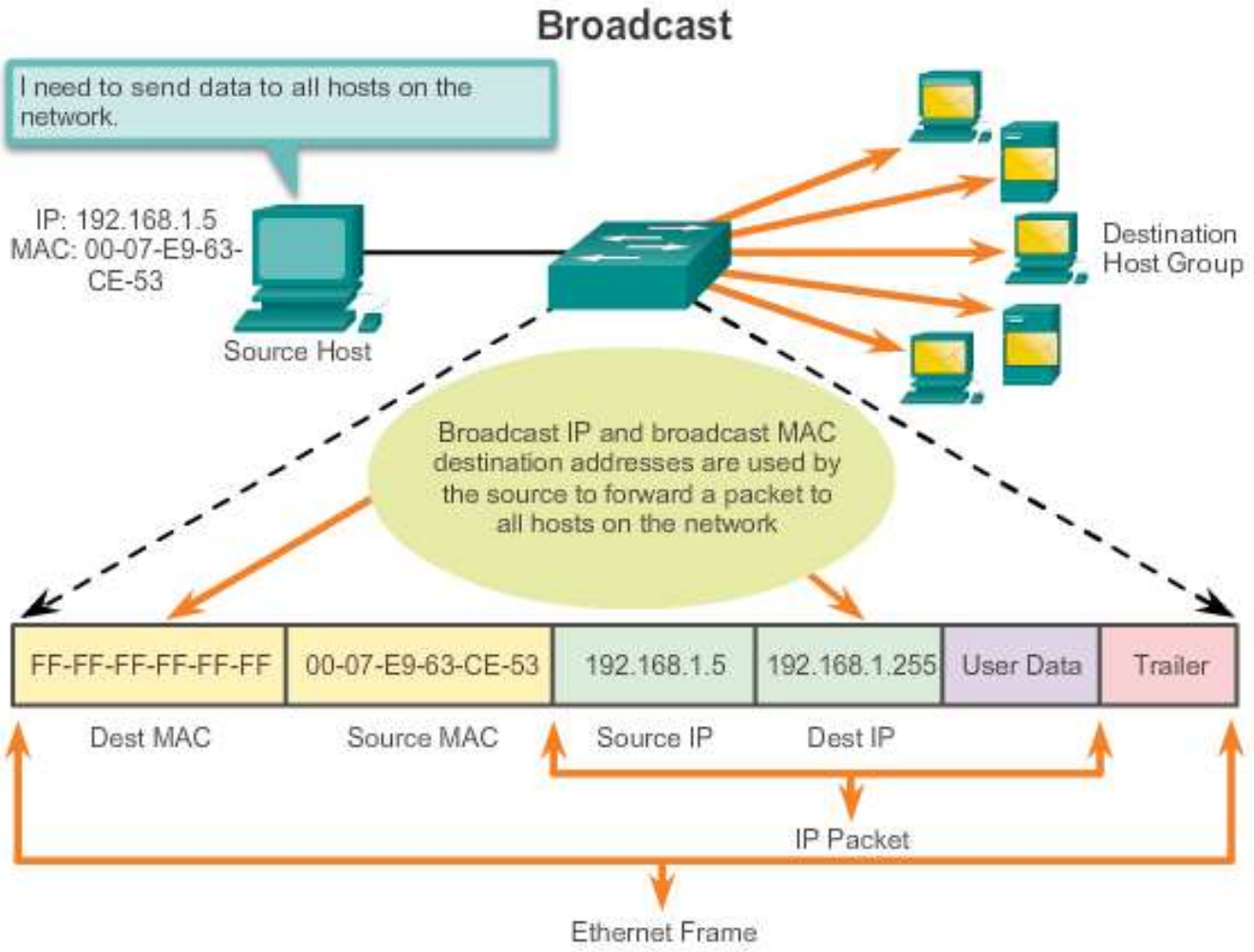
Unicast MAC Address



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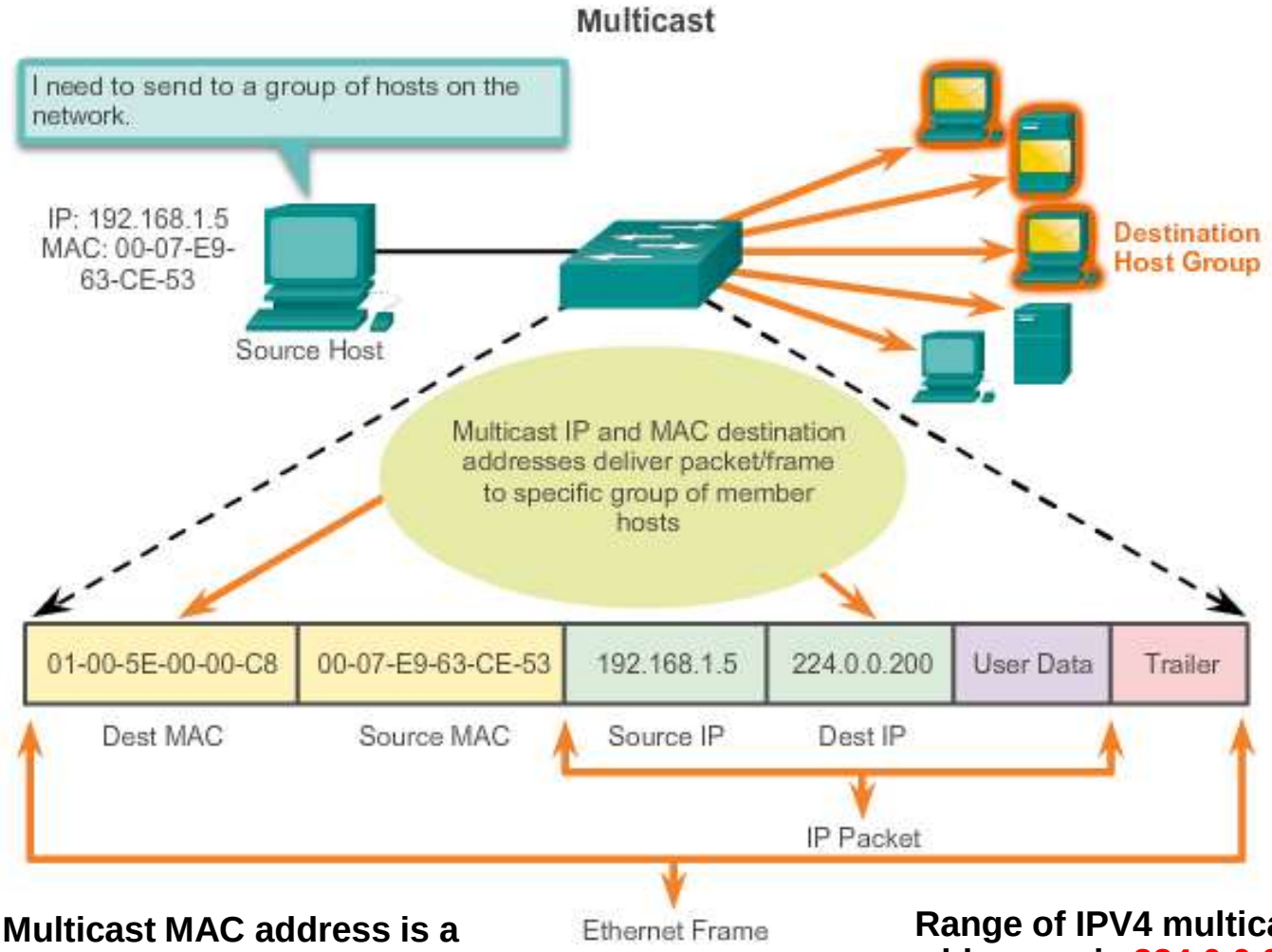
Ethernet MAC

Broadcast MAC Address



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Multicast MAC Address



Multicast MAC address is a special value that begins with 01-00-5E in hexadecimal

Range of IPV4 multicast addresses is 224.0.0.0 to 239.255.255.255

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MAC and IP

MAC Address

- This address does not change
- Similar to the name of a person
- Known as physical address because physically assigned to the host NIC

IP Address

- Similar to the address of a person
- Based on where the host is actually located
- Known as a logical address because assigned logically
- Assigned to each host by a network administrator

Both the physical MAC and logical IP addresses are required for a computer to communicate just like both the name and address of a person are required to send a letter.

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5.2 Address Resolution Protocol



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Introduction to (Address Resolution Protocol) ARP

ARP Purpose

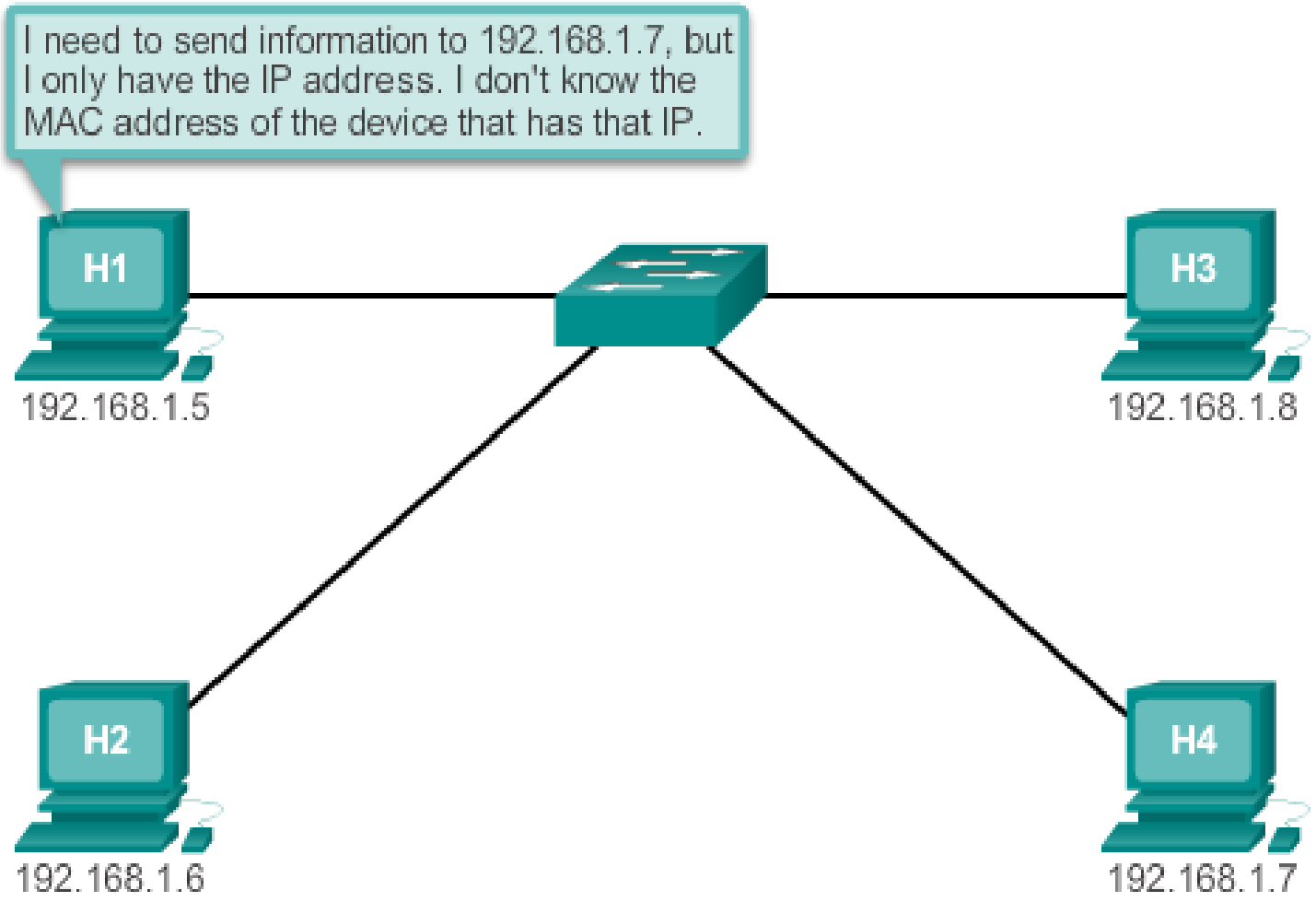
- Sending node needs a way to find the MAC address of the destination for a given Ethernet link

The ARP protocol provides two basic functions:

- Resolving IPv4 addresses to MAC addresses
- Maintaining a table of mappings

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Introduction to ARP (cont.)



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ARP Functions/Operation

ARP Table

- Used to find the data link layer address that is mapped to the destination IPv4 address.
- As a node receives frames from the media, it records the source IP and MAC address as a mapping in the ARP table.

ARP Request

- Layer 2 broadcast to all devices on the Ethernet LAN.
- The node that matches the IP address in the broadcast will reply.
- If no device responds to the ARP request, the packet is dropped because a frame cannot be created.

Note: Static map entries can be entered in an ARP table, but this is rarely done.

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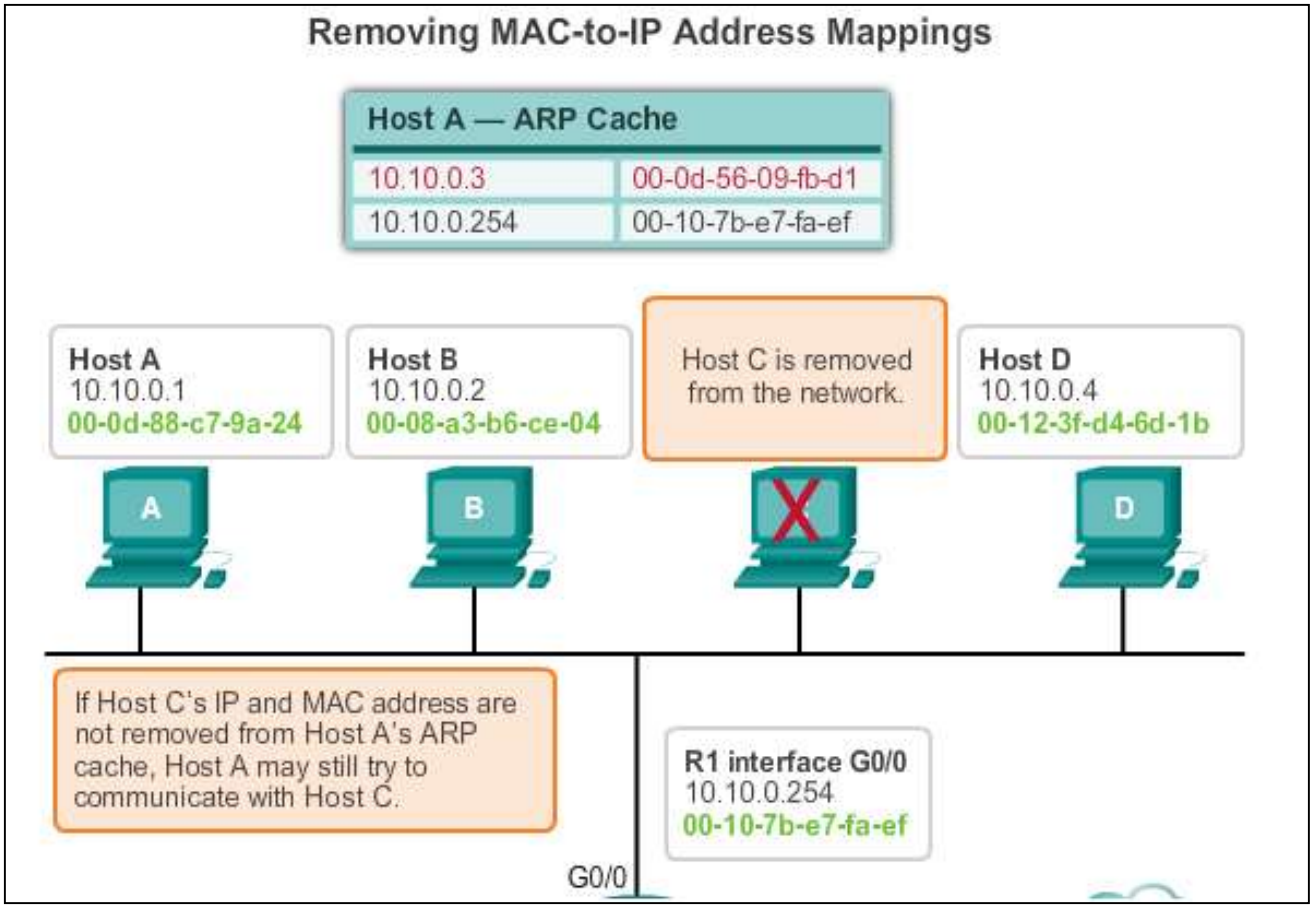
ARP Role in Remote Communication

- If the destination IPv4 host is on the local network, the frame will use the MAC address of this device as the destination MAC address.
- If the destination IPv4 host is not on the local network, the source uses the ARP process to determine a MAC address for the router interface serving as the gateway.
- In the event that the gateway entry is not in the table, an ARP request is used to retrieve the MAC address associated with the IP address of the router interface.

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Removing Entries from an ARP Table

- The ARP cache timer removes ARP entries that have not been used for a specified period of time.
- Commands may also be used to manually remove all or some of the entries in the ARP table.



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ARP Tables on Networking Devices

```
Router#show ip arp

                     Age
Protocol  Address          (min)  Hardware Addr   Type   Interface
Internet  172.16.233.229    -    0000.0c59.f892   ARPA   Ethernet0/0
Internet  172.16.233.218    -    0000.0c07.ac00   ARPA   Ethernet0/0
Internet  172.16.168.11     -    0000.0c63.1300   ARPA   Ethernet0/0
Internet  172.16.168.254    9    0000.0c36.6965   ARPA   Ethernet0/0
```

```
C:\>arp -a

Interface: 192.168.1.67 --- 0xa
  Internet Address      Physical Address      Type
  192.168.1.254         64-0f-29-0d-36-91     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```


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5.3 LAN Switches



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Switch Port Fundamentals

Layer 2 LAN Switch

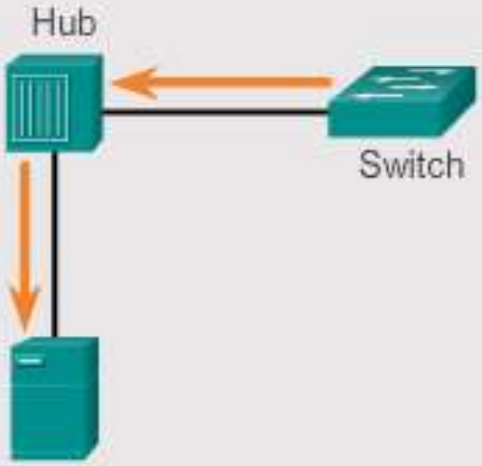
- Connects end devices to a central intermediate device on most Ethernet networks
- Performs switching and filtering based only on the MAC address
- Builds a MAC address table that it uses to make forwarding decisions
- Depends on routers to pass data between IP subnetworks

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Duplex Settings

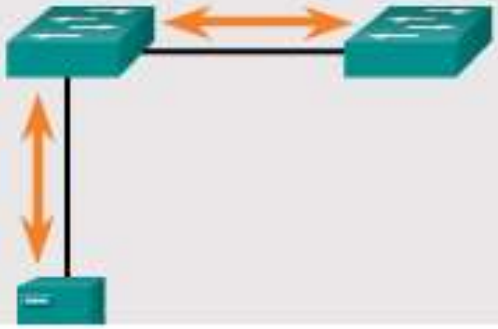
Half Duplex (CSMA/CD)

- Unidirectional data flow
- Higher potential for collision
- Hub connectivity



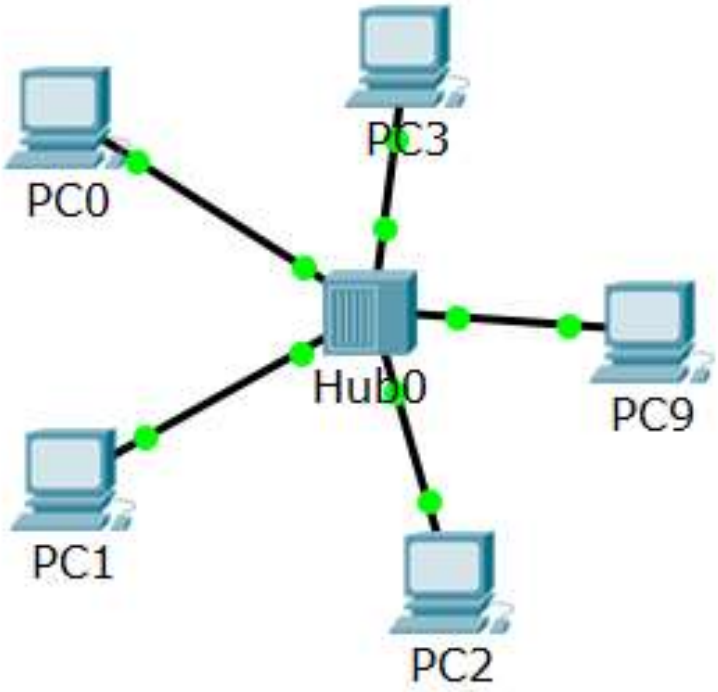
Full Duplex

- Point-to-point only
- Attached to dedicated switched port
- Requires full-duplex support on both ends
- Collision-free
- Collision detect circuit disabled

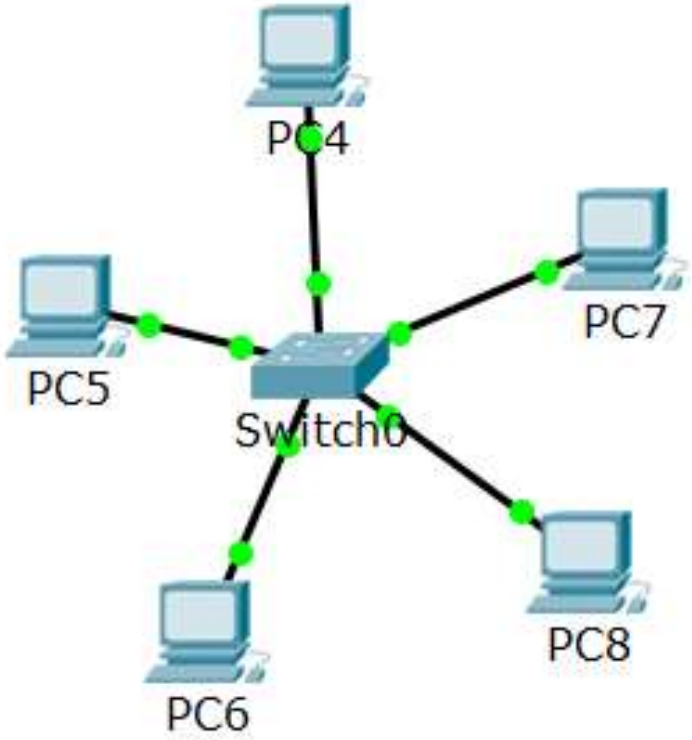


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1x Collision Domain

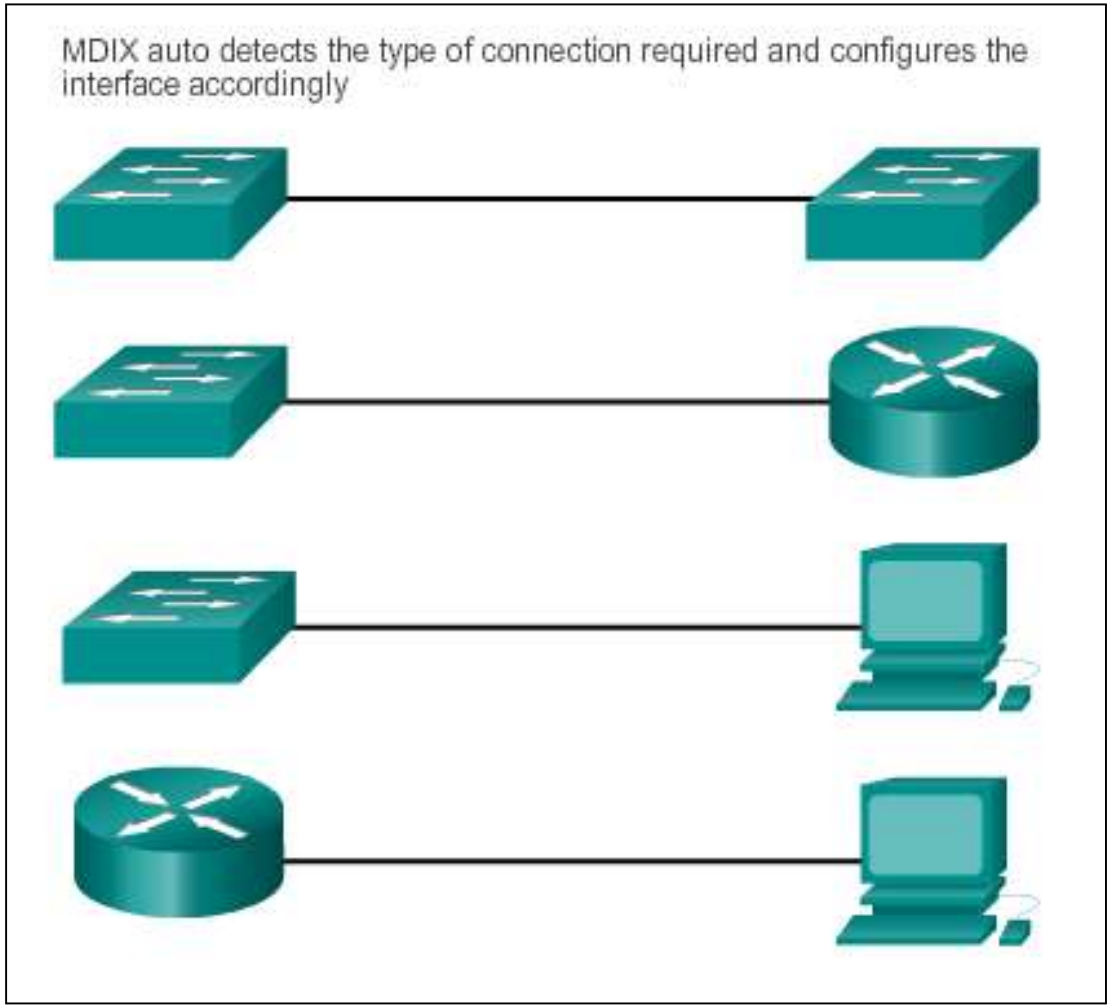


5x Collision Domains



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Auto-MDIX (Medium Dependent Interface Crossover)



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Frame Forwarding Methods on Cisco Switches

Store-and-forward



A store-and-forward switch receives the entire frame, and computes the CRC. If the CRC is valid, the switch looks up the destination address, which determines the outgoing interface. The frame is then forwarded out the correct port.

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Cut-through Switching

Fast-forward switching:

- Lowest level of latency immediately forwards a packet after reading the **destination address**, typical cut-through method of switching

Fragment-free switching:

- Switch stores the **first 64** bytes of the frame before forwarding, most network errors and collisions occur during the first 64 bytes



A cut-through switch forwards the frame before it is entirely received. At a minimum, the destination address of the frame must be read before the frame can be forwarded.

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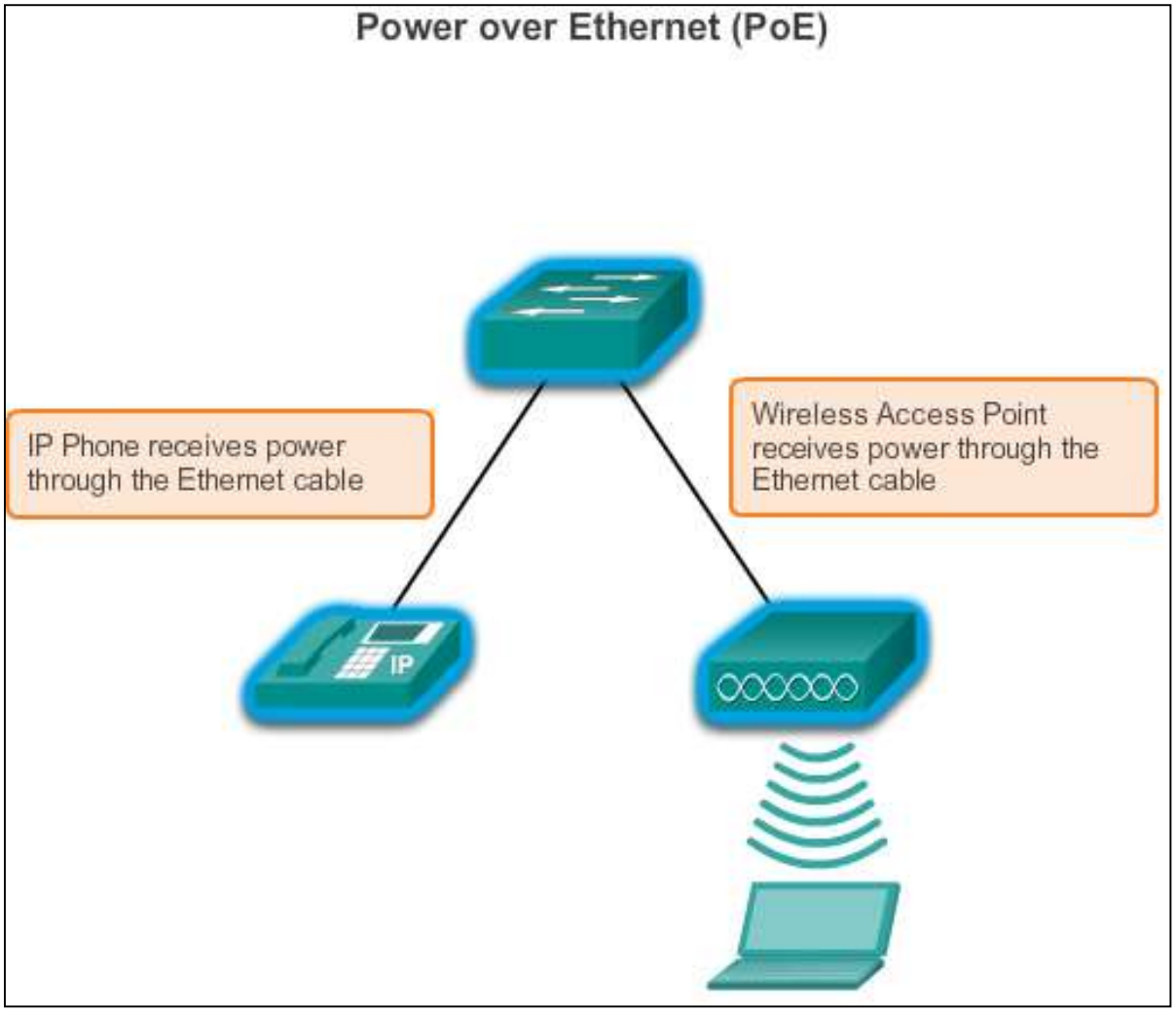
Memory Buffering on Switches

Port-Based and Shared Memory Buffering

Port-based memory	In port-based memory buffering, frames are stored in queues that are linked to specific incoming and outgoing ports.
Shared memory	Shared memory buffering deposits all frames into a common memory buffer, which all the ports on the switch share.

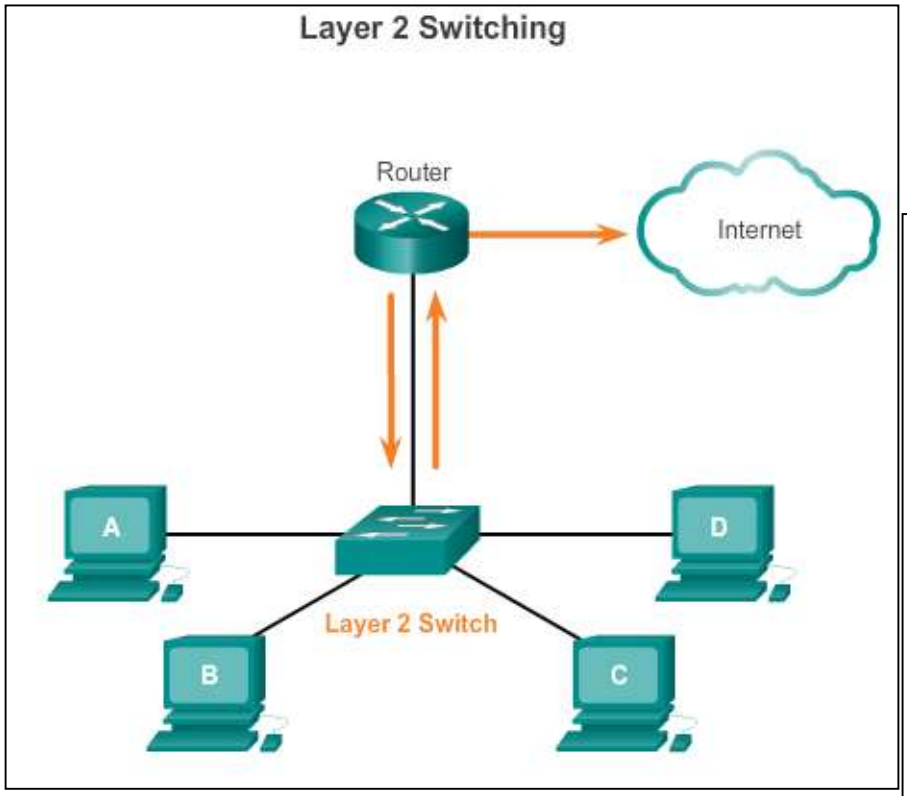
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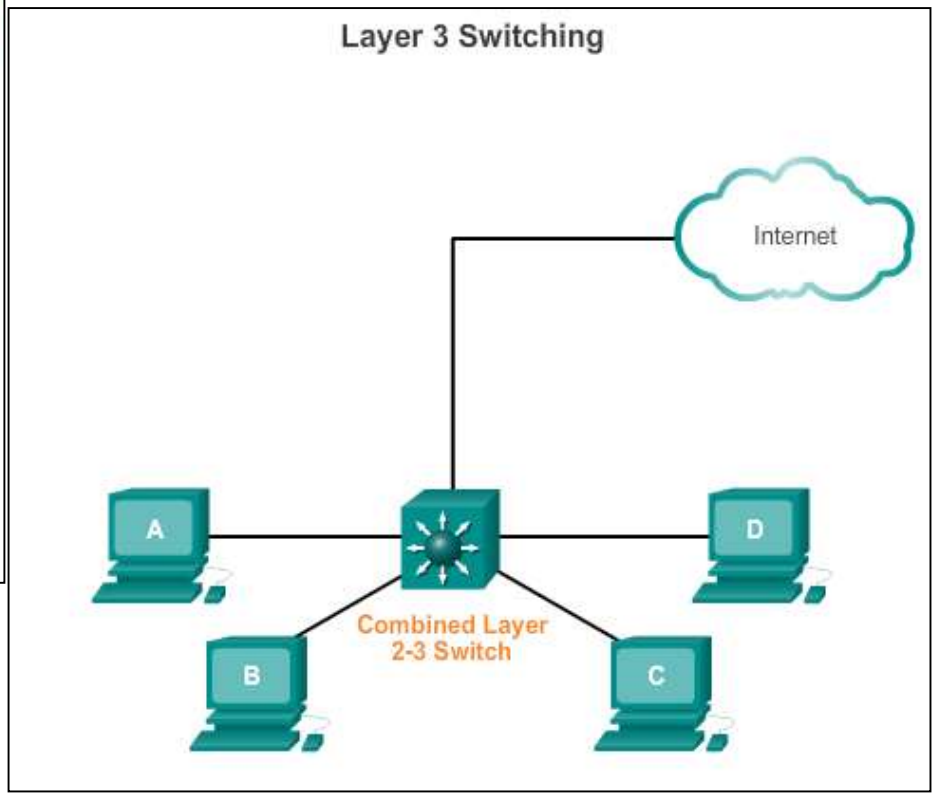
Fixed versus Modular Configuration



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Layer 2 versus Layer 3 Switching





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Types of Layer 3 Interfaces

The major types of Layer 3 interfaces are:

- **Switch Virtual Interface (SVI)** – Logical interface on a switch associated with a virtual local-area network (VLAN).
- **Routed Port** – Physical port on a Layer 3 switch configured to act as a router port. Configure routed ports by putting the interface into Layer 3 mode with the **no switchport** interface configuration command.
- **Layer 3 EtherChannel** – Logical interface on a Cisco device associated with a *bundle* of routed ports (port aggregation).

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Configuring a Routed Port on a Layer 3 Switch

Routed Port Configuration

```
S1(config)#interface f0/6
S1(config-if)#no switchport
S1(config-if)#ip address 192.168.200.1 255.255.255.0
S1(config-if)#no shutdown
S1(config-if)#end
S1#
*Mar  1 00:15:40.115: %SYS-5-CONFIG_I: Configured from console by console
S1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
Vlan1                  unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  down                  down
FastEthernet0/2        unassigned      YES unset  down                  down
FastEthernet0/3        unassigned      YES unset  down                  down
FastEthernet0/4        unassigned      YES unset  down                  down
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        192.168.200.1   YES manual up                    up
FastEthernet0/7        unassigned      YES unset  up                    up
FastEthernet0/8        unassigned      YES unset  up                    up
<output omitted>
```


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Summary

- Ethernet is the most widely used LAN technology used today.
- Ethernet standards define both the Layer 2 protocols and the Layer 1 technologies.
- The Ethernet frame structure adds headers and trailers around the Layer 3 PDU to encapsulate the message being sent.
- As an implementation of the IEEE 802.2/3 standards, the Ethernet frame provides MAC addressing and error checking.
- Replacing hubs with switches in the local network has reduced the probability of frame collisions in half-duplex links.
- The Layer 2 addressing provided by Ethernet supports unicast, multicast, and broadcast communications.
- Ethernet uses the Address Resolution Protocol to determine the MAC addresses of destinations and map them against known Network layer addresses.

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